

Flight Director Gene Kranz told his team at the Manned Spacecraft Center in Houston: “From this day forward, Flight Control will be known by two words: ‘tough’ and ‘competent.’ ‘Tough’ means we are forever accountable for what we do or what we fail to do. ‘Competent’ means we will never take anything for granted.”

The Apollo program was put on hold while an exhaustive investigation was made of the accident. It was concluded that the most likely cause was a spark from a short circuit in a bundle of wires that ran to the left and just in front of Grissom’s seat. The large amount of flammable material in the cabin in the oxygen environment allowed the fire to start and spread quickly. A number of changes were instigated in the program over the next year and a half, including designing a new hatch which opened outward and could be operated quickly, removing much of the flammable material and replacing it with self-extinguishing components, using a nitrogen-oxygen mixture at launch, and recording all changes and overseeing all modifications to the spacecraft design more rigorously.

Apollo 2-10

Following the tragedy, NASA grew more cautious and went back to the basics. Two test rockets were fired without any personnel on board, just to check the abilities of the Saturn I launch vehicle. These were unofficially dubbed the Apollo 2 and Apollo 3.

Apollo 4 was the first unmanned mission in the programme, and used the modified Saturn V launch vehicle. It was fitted with a complete Command/Service Module and on November 9, 1967, it successfully blasted off from Cape Canaveral. The mission provided vital information on the need for dampening mechanisms at the launchpad to protect structures in its vicinity.

Thereafter, Apollo 5 was launched on January 22, 1968, and became the first test of the Lunar Module, followed quickly by the Apollo 6 unmanned flight which made the scientists confident enough to put astronauts in the next mission.

Apollo 7 would become the first manned mission since the tragic loss of Apollo 1. On Oct 11, 1968, the rocket took off carrying Walter Schirra Jr., Walter Cunningham and Donn Eisele. At five minutes, 54 seconds into the mission, Schirra Jr., the commander, reported, “She is riding like a dream.” About ten minutes after launch, Apollo 7 reached the first stage of its journey, an orbital path 227 by 285 kilometers above Earth. The Apollo vehicle and the CSM performed superbly. Durability was shown for 10.8